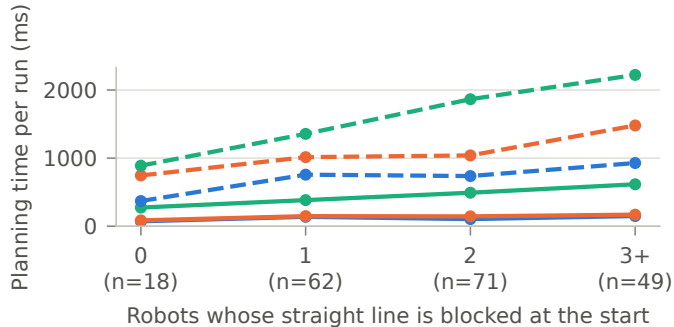
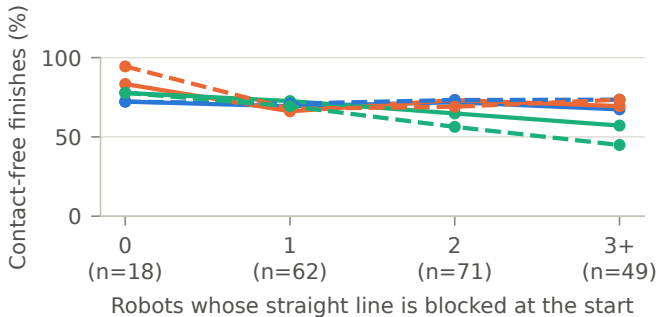


Safety and planning time vs crowding for three planners

Visibility Graph loses the most as more robots start blocked (contact-free 78 % $\rightarrow$ 57 % with event, 78 % $\rightarrow$ 45 % with cycle), while Voronoi and PRM stay roughly level. Scenarios grouped by how many robots need a detour at the start (straight line blocked for all three planners); n = scenarios per group. Solid lines: event (ours); dashed: cycle; 20 ms checks, no prediction.



Voronoi PRM Visibility Graph Event (ours) @ 20 ms Cycle @ 20 ms